

## UBERLANDIA, Brazil

WMO: 835250

Lat: 18.883S

Lon: 48.220W

Elev: 943

StdP: 90.50

Time Zone: -3.00 (SAB)

Period: 98-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |     |     |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD<br>to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|-----|-----|------|-----------------------|------|-----|------|--------------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99% |     |      | 0.4%                  |      | 1%  |      |                          |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP  | HR  | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                     | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g) | (h) | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                      | (o)  | (p)   |
| 6             | 11.0       | 12.9 | 0.9                           | 4.5 | 23.3 | 2.8 | 5.2 | 24.0 | 8.5                   | 21.4 | 7.9 | 21.2 | 4.8                      | 100  | 0.356 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 10            | 10.4                   | 32.2            | 17.7 | 31.2 | 18.2 | 30.2 | 18.8 | 22.9                | 26.8 | 22.4 | 26.5 | 22.0 | 26.3 | 4.0                  | 30   |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 22.0                            | 18.7 | 24.3 | 21.2 | 17.8 | 23.5 | 21.0 | 17.6 | 23.3 | 73.0          | 26.5 | 71.1 | 26.0 | 69.5 | 26.4 | 26.9              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |     |                    |     | n-Year Return Period Values of Extreme Temperature |     |            |     |            |     |            |      |      |  |
|-------------------|------|-----|----------------------------|-----|--------------------|-----|----------------------------------------------------|-----|------------|-----|------------|-----|------------|------|------|--|
| 1%                | 2.5% | 5%  | Mean                       |     | Standard Deviation |     | n=5 years                                          |     | n=10 years |     | n=20 years |     | n=50 years |      |      |  |
|                   |      |     | Min                        | Max | Min                | Max | Min                                                | Max | Min        | Max | Min        | Max | Min        | Max  |      |  |
| (a)               | (b)  | (c) | (e)                        | (f) | (g)                | (h) | (i)                                                | (j) | (k)        | (l) | (m)        | (n) | (o)        | (p)  |      |  |
| 8.5               | 7.6  | 6.9 |                            |     |                    |     |                                                    |     |            |     |            |     |            |      |      |  |
|                   |      |     | DB                         | 6.7 | 34.2               | 1.6 | 1.4                                                | 5.5 | 35.2       | 4.6 | 36.0       | 3.7 | 36.8       | 2.6  | 37.8 |  |
|                   |      |     | WB                         | 5.1 | 23.2               | 2.0 | 0.9                                                | 3.6 | 23.8       | 2.4 | 24.4       | 1.3 | 24.9       | -0.2 | 25.6 |  |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 22.5   | 23.5  | 23.9  | 23.5  | 22.7  | 20.7  | 20.0  | 19.9  | 21.6  | 23.7  | 24.2  | 23.4  | 23.4  |
|                                                                                  | DBStd       | 2.36   | 1.52  | 1.26  | 1.31  | 1.61  | 1.79  | 1.72  | 2.17  | 2.05  | 2.50  | 2.41  | 1.54  | 1.44  |
|                                                                                  | HDD10.0     | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
|                                                                                  | HDD18.3     | 25     | 0     | 0     | 0     | 0     | 4     | 6     | 11    | 3     | 1     | 1     | 0     | 0     |
|                                                                                  | CDD10.0     | 4573   | 417   | 389   | 417   | 381   | 332   | 301   | 308   | 359   | 411   | 442   | 402   | 414   |
|                                                                                  | CDD18.3     | 1556   | 159   | 156   | 159   | 131   | 78    | 56    | 61    | 103   | 162   | 184   | 152   | 156   |
|                                                                                  | CDH23.3     | 9792   | 876   | 925   | 832   | 688   | 371   | 271   | 378   | 789   | 1443  | 1503  | 893   | 824   |
|                                                                                  | CDH26.7     | 2524   | 214   | 234   | 178   | 117   | 27    | 11    | 34    | 177   | 550   | 576   | 223   | 182   |
| Wind                                                                             | WSAvg       | 3.7    | 3.5   | 3.3   | 3.3   | 3.4   | 3.5   | 3.7   | 3.9   | 4.2   | 4.4   | 4.1   | 3.8   | 3.5   |
| Precipitation                                                                    | PrecAvg     | 1400   | 279   | 205   | 208   | 65    | 37    | 13    | 6     | 14    | 49    | 95    | 180   | 252   |
|                                                                                  | PrecMax     | 1637   | 472   | 367   | 360   | 130   | 108   | 96    | 25    | 46    | 102   | 180   | 266   | 379   |
|                                                                                  | PrecMin     | 1185   | 118   | 42    | 89    | 18    | 1     | 0     | 0     | 0     | 0     | 42    | 101   | 164   |
|                                                                                  | PrecStd     | 140    | 99    | 97    | 78    | 31    | 26    | 24    | 8     | 16    | 30    | 38    | 48    | 56    |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 31.1  | 31.2  | 30.2  | 30.2  | 28.2  | 27.9  | 28.9  | 31.1  | 33.9  | 34.9  | 31.8  | 30.9  |
|                                                                                  |             | MCWB   | 19.7  | 20.8  | 21.1  | 20.9  | 18.3  | 17.0  | 15.8  | 15.9  | 16.8  | 17.7  | 19.9  | 20.3  |
|                                                                                  | 2%          | DB     | 30.0  | 30.1  | 29.8  | 29.1  | 27.2  | 26.9  | 27.8  | 29.9  | 32.2  | 32.9  | 30.1  | 29.9  |
|                                                                                  |             | MCWB   | 20.2  | 20.7  | 21.1  | 20.4  | 18.0  | 17.0  | 15.9  | 15.6  | 16.8  | 17.9  | 20.0  | 20.4  |
|                                                                                  | 5%          | DB     | 29.0  | 29.1  | 28.9  | 28.0  | 26.2  | 26.0  | 26.9  | 28.8  | 31.2  | 31.2  | 29.1  | 28.8  |
|                                                                                  |             | MCWB   | 20.5  | 20.6  | 21.0  | 20.1  | 18.0  | 16.9  | 15.7  | 15.7  | 16.6  | 18.2  | 20.2  | 20.5  |
|                                                                                  | 10%         | DB     | 27.9  | 28.1  | 27.8  | 27.0  | 25.8  | 25.0  | 25.8  | 27.2  | 30.0  | 30.0  | 28.0  | 27.8  |
|                                                                                  |             | MCWB   | 20.7  | 20.8  | 20.9  | 19.9  | 17.9  | 16.7  | 15.6  | 15.4  | 16.6  | 18.4  | 20.3  | 20.7  |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 23.2  | 23.6  | 23.1  | 22.2  | 20.7  | 19.7  | 19.0  | 19.2  | 21.5  | 22.1  | 23.0  | 23.5  |
|                                                                                  |             | MCDB   | 26.6  | 27.6  | 27.7  | 27.5  | 25.2  | 24.6  | 24.4  | 24.8  | 26.6  | 27.6  | 26.9  | 26.3  |
|                                                                                  | 2%          | WB     | 22.5  | 22.7  | 22.4  | 21.6  | 20.1  | 18.9  | 17.8  | 17.9  | 20.4  | 21.4  | 22.1  | 22.6  |
|                                                                                  |             | MCDB   | 26.2  | 26.8  | 27.0  | 26.6  | 24.7  | 24.2  | 24.1  | 24.5  | 24.9  | 26.6  | 26.5  | 26.0  |
|                                                                                  | 5%          | WB     | 22.0  | 22.1  | 22.0  | 21.2  | 19.4  | 18.3  | 17.1  | 17.2  | 19.5  | 20.7  | 21.6  | 22.1  |
|                                                                                  |             | MCDB   | 25.9  | 26.4  | 26.5  | 25.8  | 23.8  | 23.6  | 23.5  | 24.3  | 24.6  | 25.8  | 26.0  | 25.7  |
|                                                                                  | 10%         | WB     | 21.5  | 21.6  | 21.5  | 20.7  | 18.7  | 17.6  | 16.5  | 16.5  | 18.7  | 20.3  | 21.1  | 21.6  |
|                                                                                  |             | MCDB   | 25.2  | 25.8  | 25.8  | 24.9  | 23.2  | 22.8  | 23.1  | 24.1  | 24.7  | 25.4  | 25.3  | 25.2  |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 8.4   | 8.9   | 8.3   | 9.1   | 9.8   | 10.4  | 11.0  | 11.8  | 11.6  | 10.4  | 9.1   | 8.4   |
|                                                                                  |             | MCDBR  | 10.1  | 10.2  | 9.7   | 9.9   | 10.3  | 10.8  | 11.7  | 12.4  | 12.6  | 11.9  | 10.6  | 10.1  |
|                                                                                  |             | MCWBR  | 3.5   | 3.5   | 3.5   | 3.7   | 4.1   | 4.2   | 4.3   | 4.2   | 4.1   | 3.6   | 3.6   | 3.6   |
|                                                                                  | 5% WB       | MCDBR  | 8.3   | 8.9   | 8.6   | 8.8   | 8.9   | 9.6   | 10.7  | 11.3  | 10.4  | 10.0  | 9.2   | 8.3   |
|                                                                                  |             | MCWBR  | 3.6   | 3.7   | 3.8   | 3.6   | 3.9   | 4.3   | 4.5   | 4.4   | 4.0   | 3.8   | 3.7   | 3.8   |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        |        | 0.374 | 0.374 | 0.363 | 0.350 | 0.329 | 0.321 | 0.333 | 0.360 | 0.448 | 0.426 | 0.391 | 0.382 |
|                                                                                  | taud        |        | 2.496 | 2.481 | 2.529 | 2.532 | 2.550 | 2.578 | 2.507 | 2.408 | 2.172 | 2.291 | 2.412 | 2.460 |
|                                                                                  | Ebn at Noon |        | 970   | 960   | 946   | 921   | 905   | 896   | 892   | 897   | 852   | 900   | 947   | 962   |
|                                                                                  | Edh at Noon |        | 116   | 116   | 107   | 99    | 91    | 85    | 94    | 111   | 150   | 139   | 126   | 120   |
| All-Sky Solar<br>Radiation                                                       | RadAvg      |        | 5.72  | 5.96  | 5.41  | 5.22  | 4.65  | 4.40  | 4.66  | 5.43  | 5.70  | 5.91  | 5.60  | 5.72  |
|                                                                                  | RadStd      |        | 0.68  | 0.49  | 0.47  | 0.34  | 0.21  | 0.18  | 0.22  | 0.28  | 0.48  | 0.48  | 0.32  | 0.42  |

## Historical Trends

|                       | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|-----------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                       |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                       | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| Station Only          | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| Regional (1 neighbor) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page